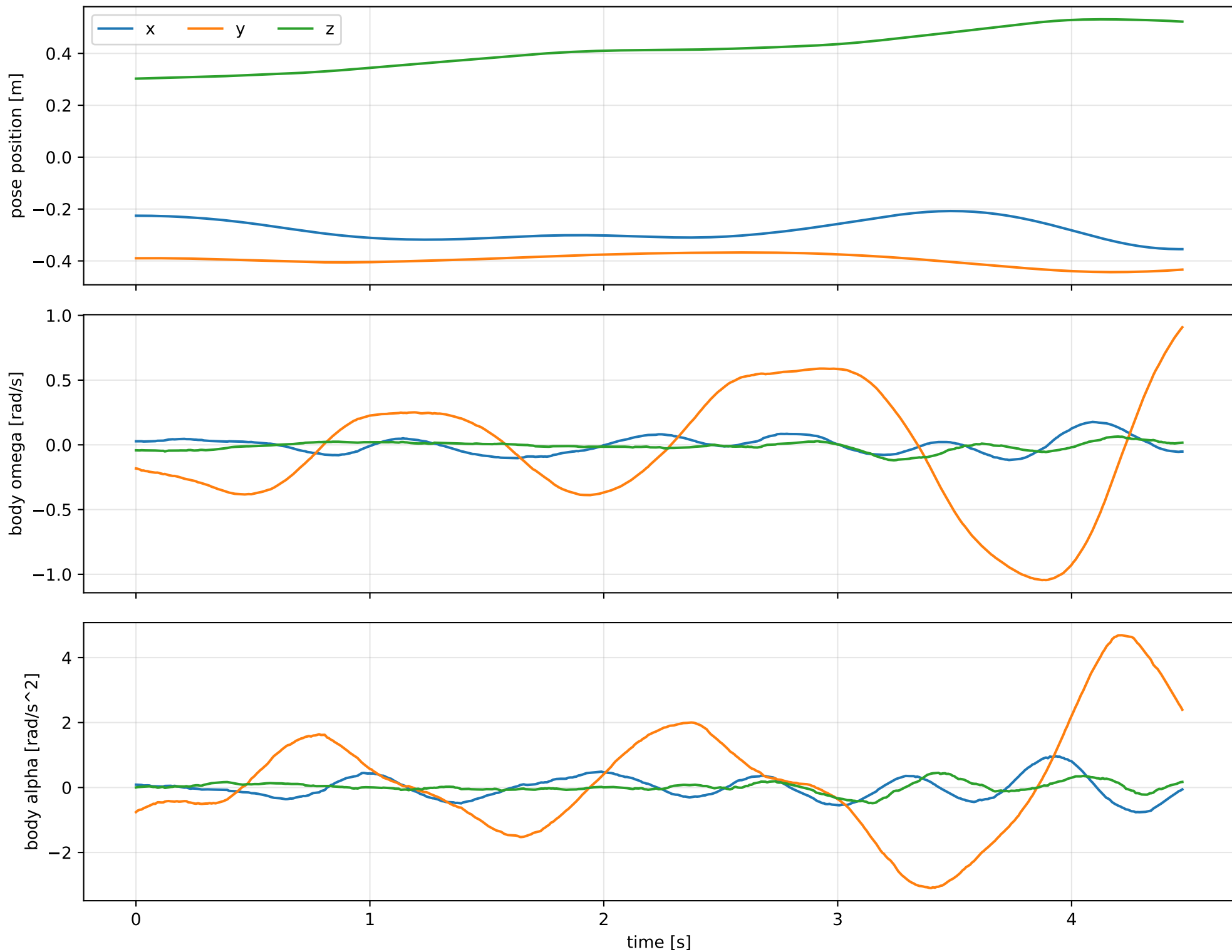
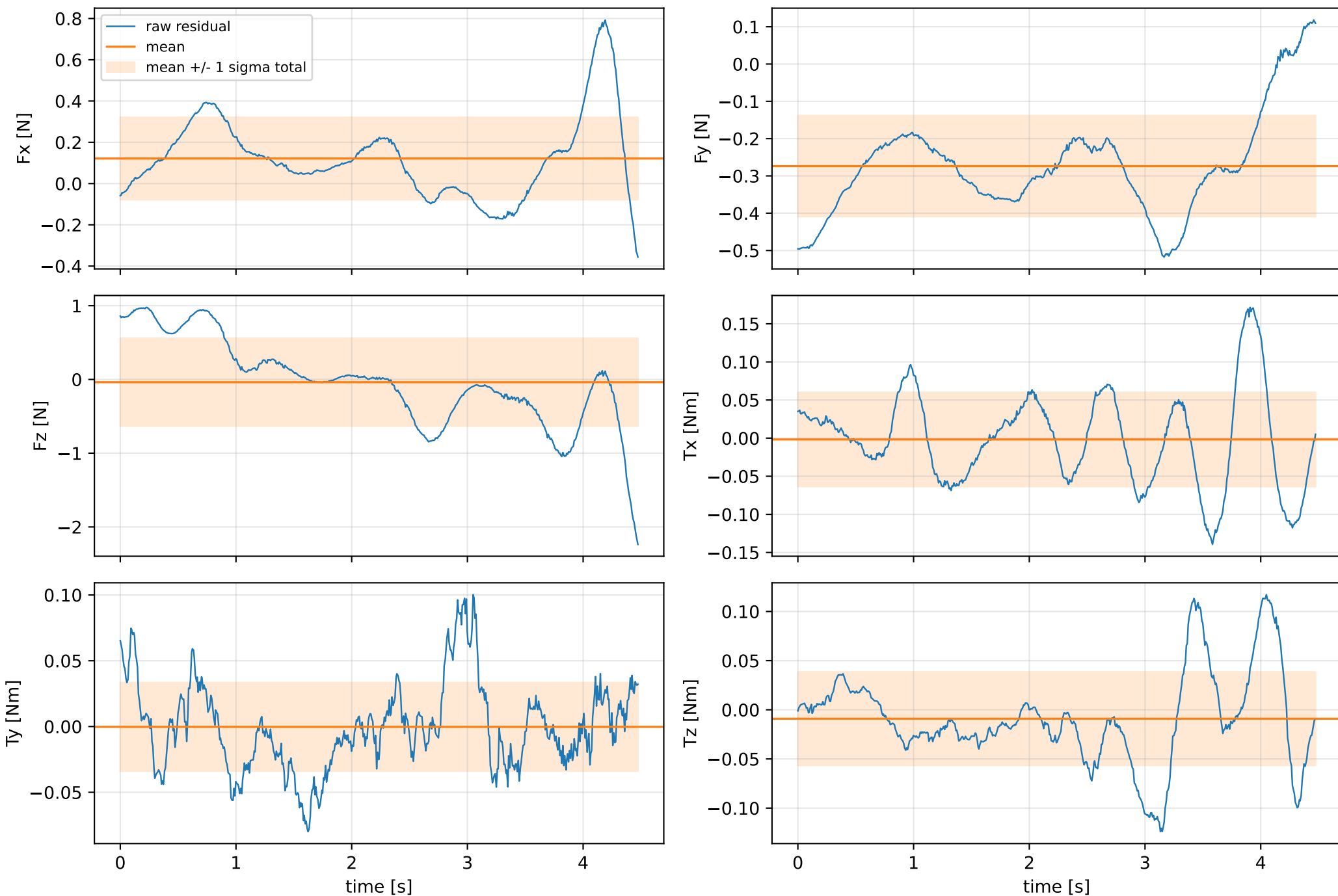


inertia_over_mass_yz_nominal: SG trajectory and derived motion



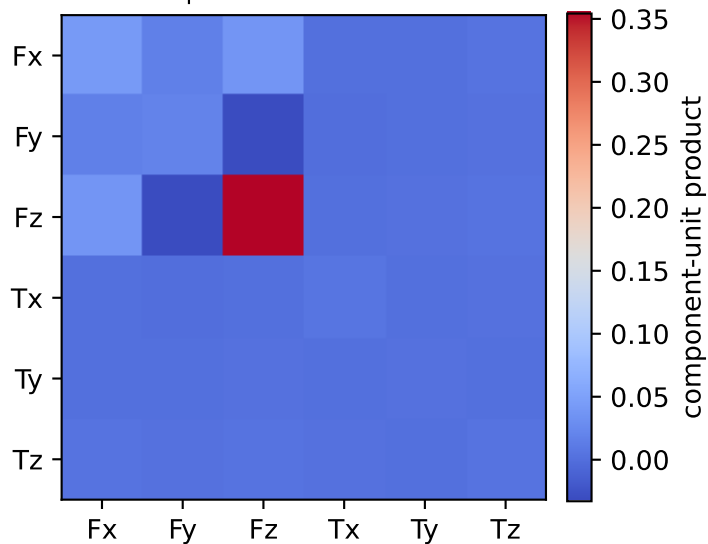
inertia_over_mass_yz_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



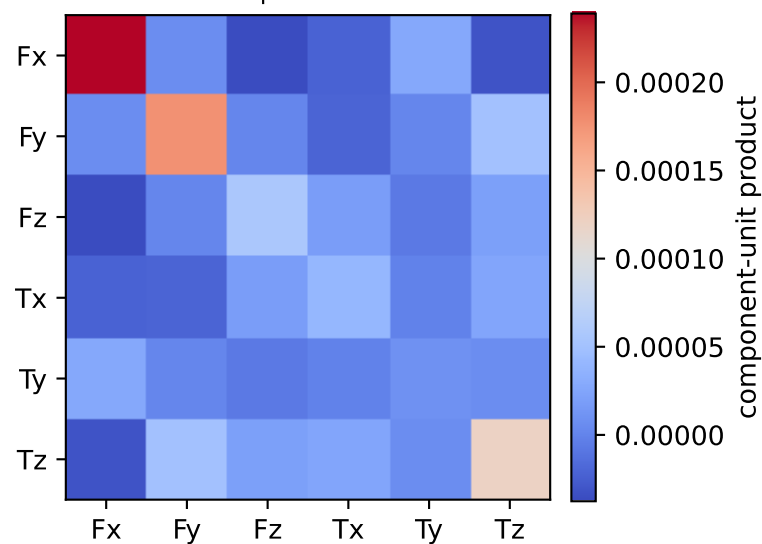
vector RMS about zero: force 0.70901 N, torque 0.085212 Nm; component std about mean: force [0.1994 0.1354 0.5952], torque [0.0617 0.0336 0.0475]

inertia_over_mass_yz_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.001 0.002 0.003 0.005 0.047 0.361]

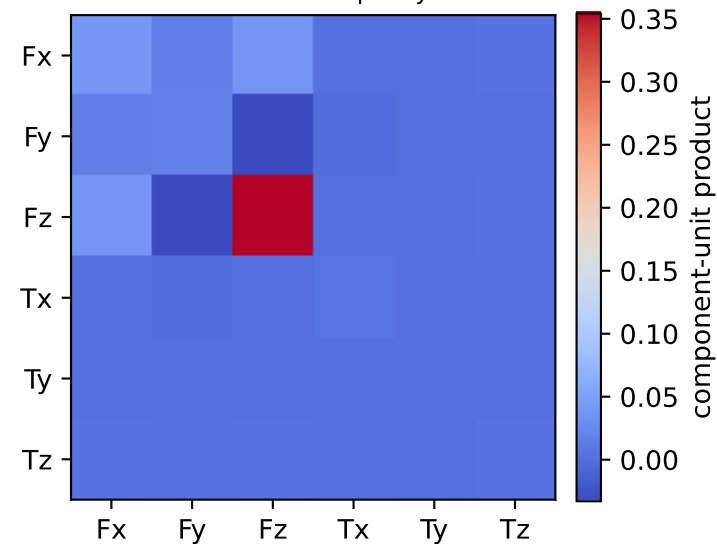
empirical total covariance



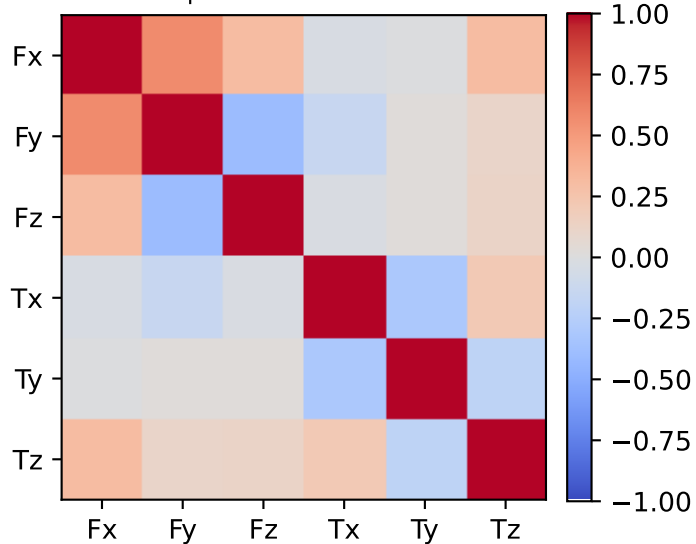
mean full-SG prediction covariance



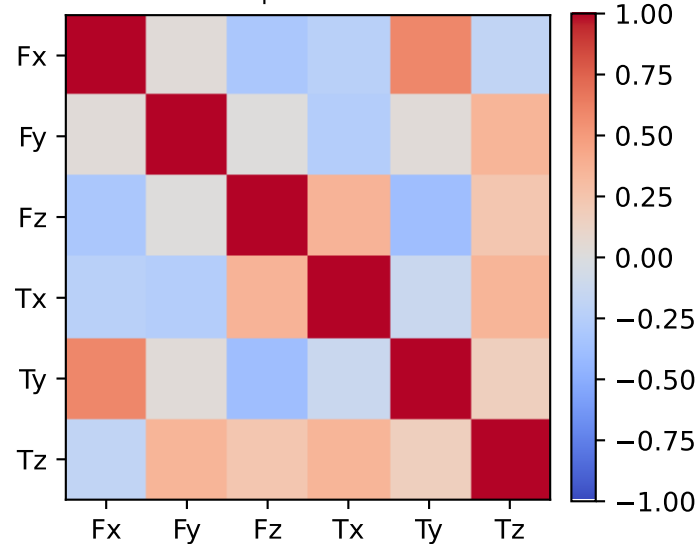
PSD excess model discrepancy covariance



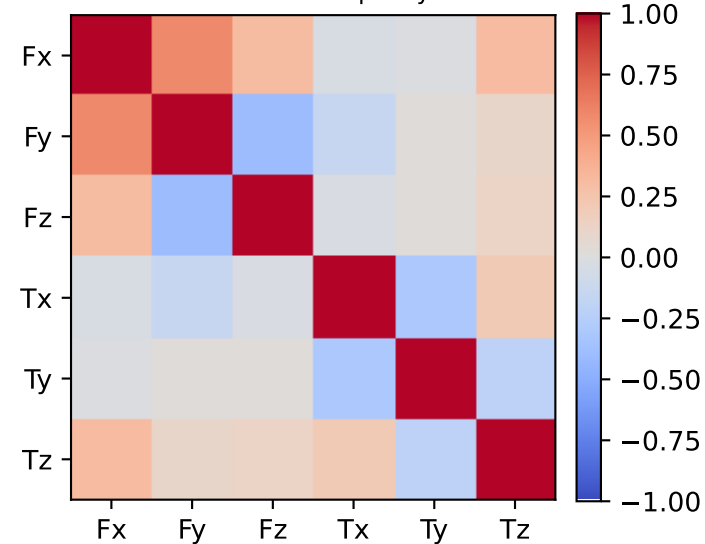
empirical total correlation



mean full-SG prediction correlation

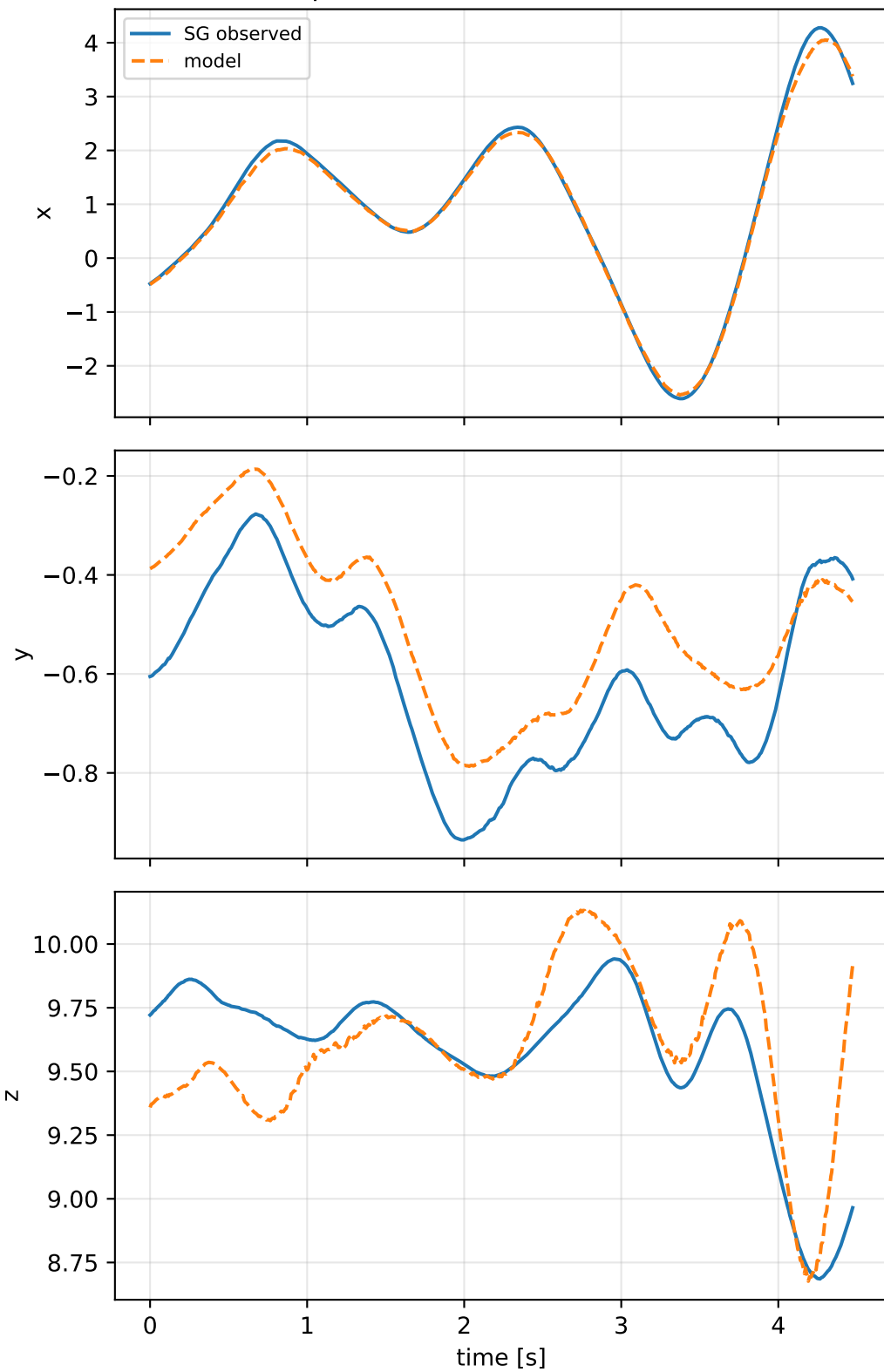


PSD excess model discrepancy correlation

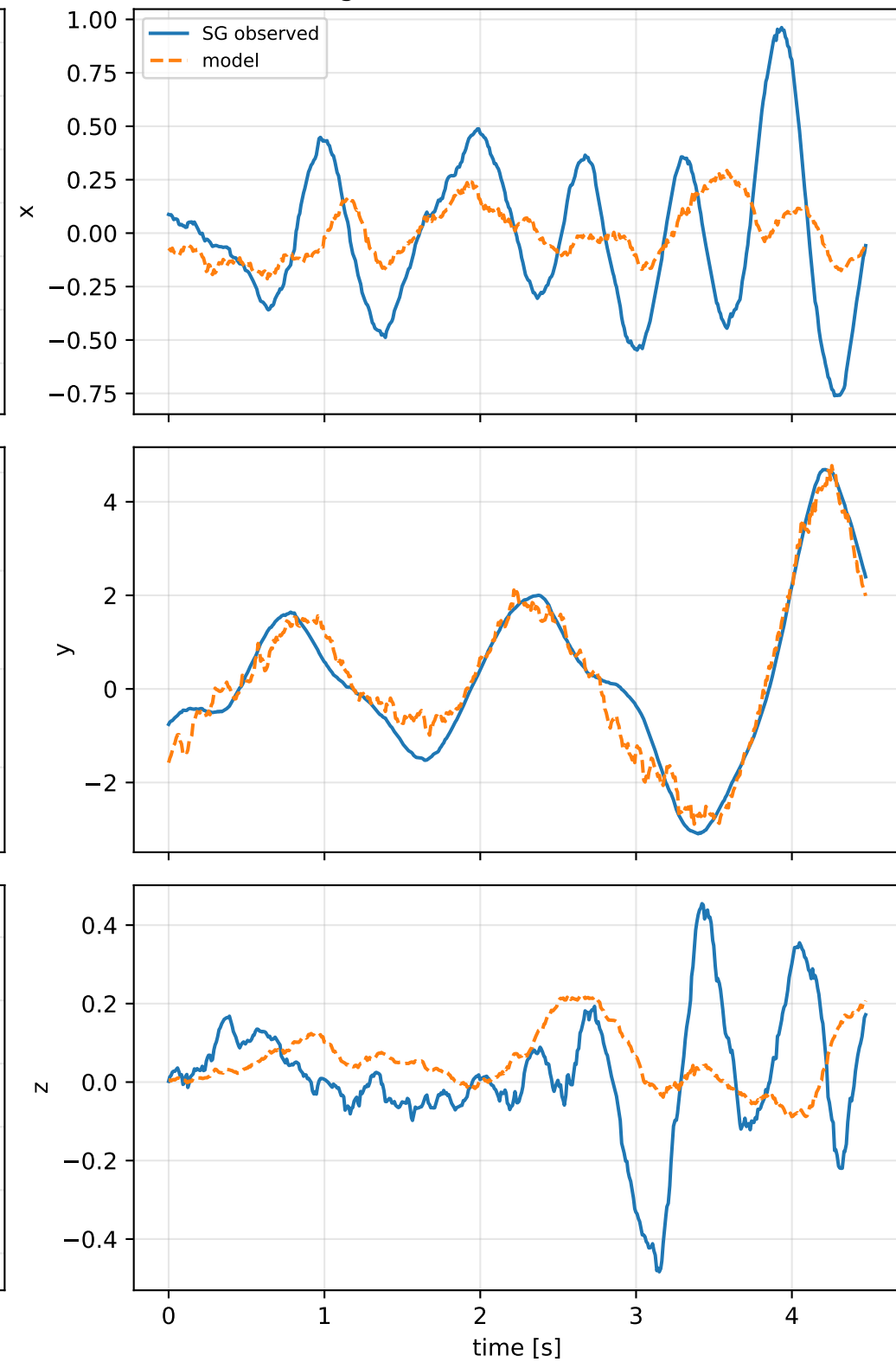


inertia_over_mass_yz_nominal: acceleration residual objective

specific acceleration [m/s^2]

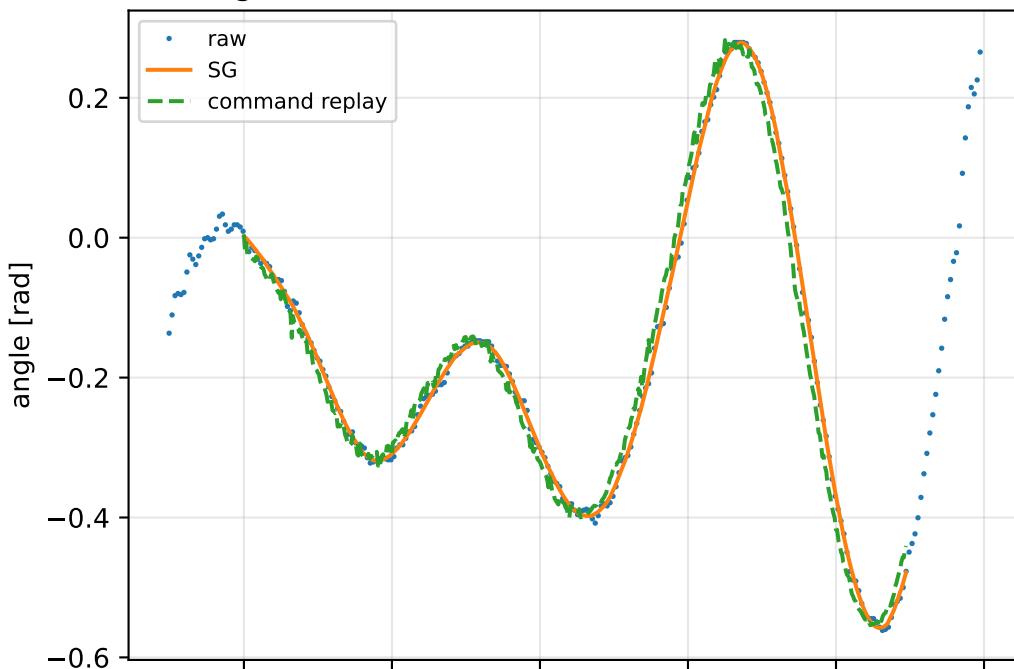


angular acceleration [rad/s^2]

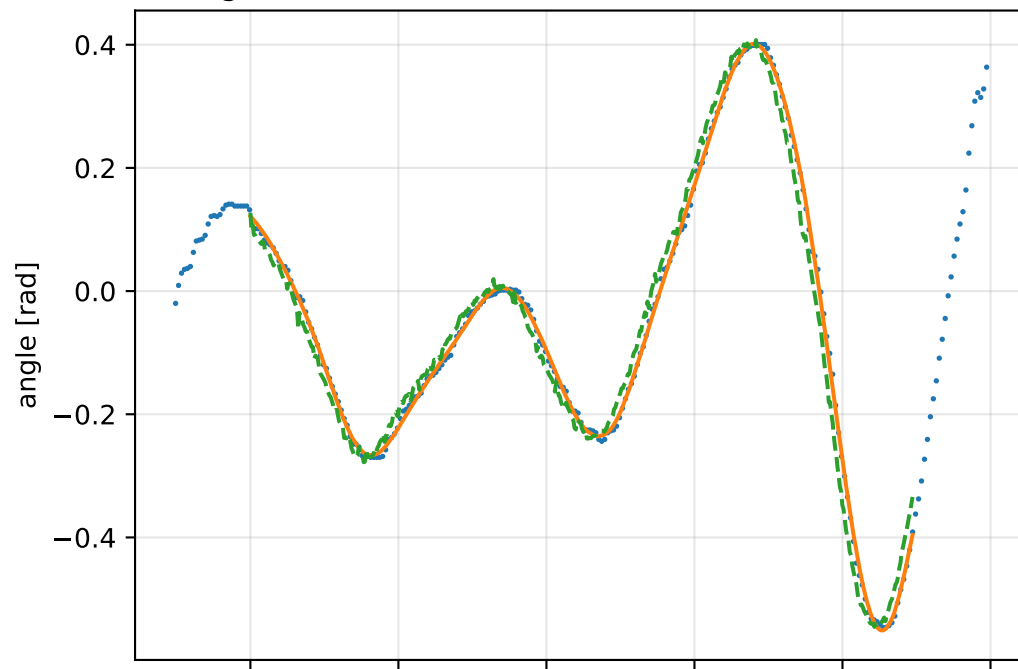


inertia_over_mass_yz_nominal: gimbal measurement audit (objective=measured_sg)

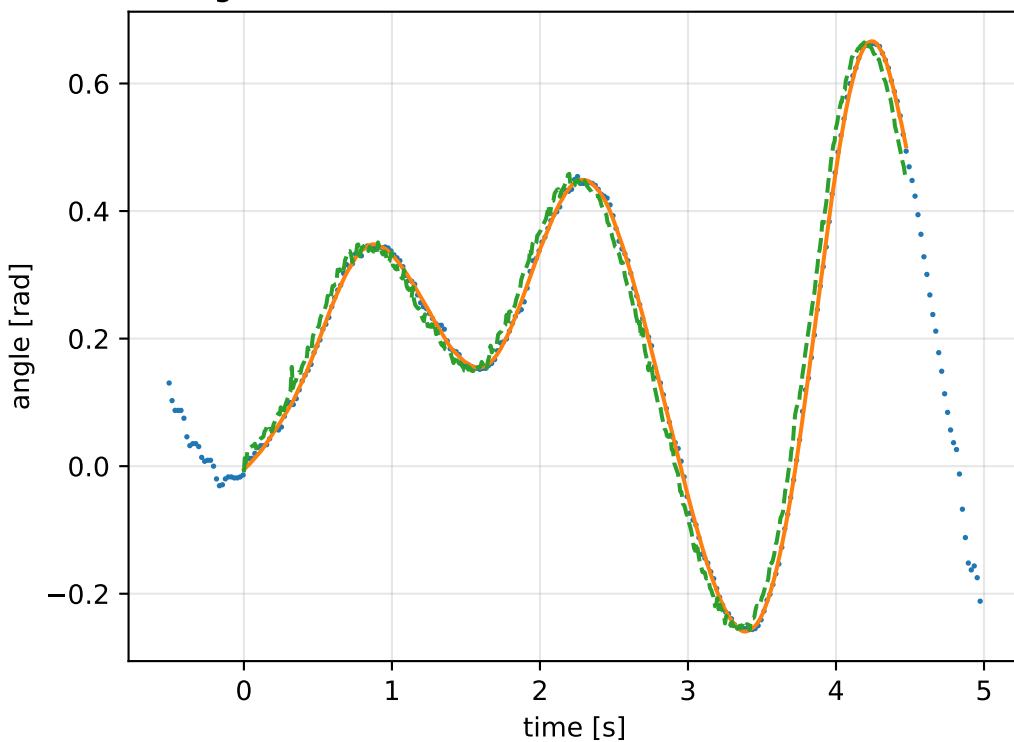
gimbal 1: RMSE=0.027, max=0.07529 rad



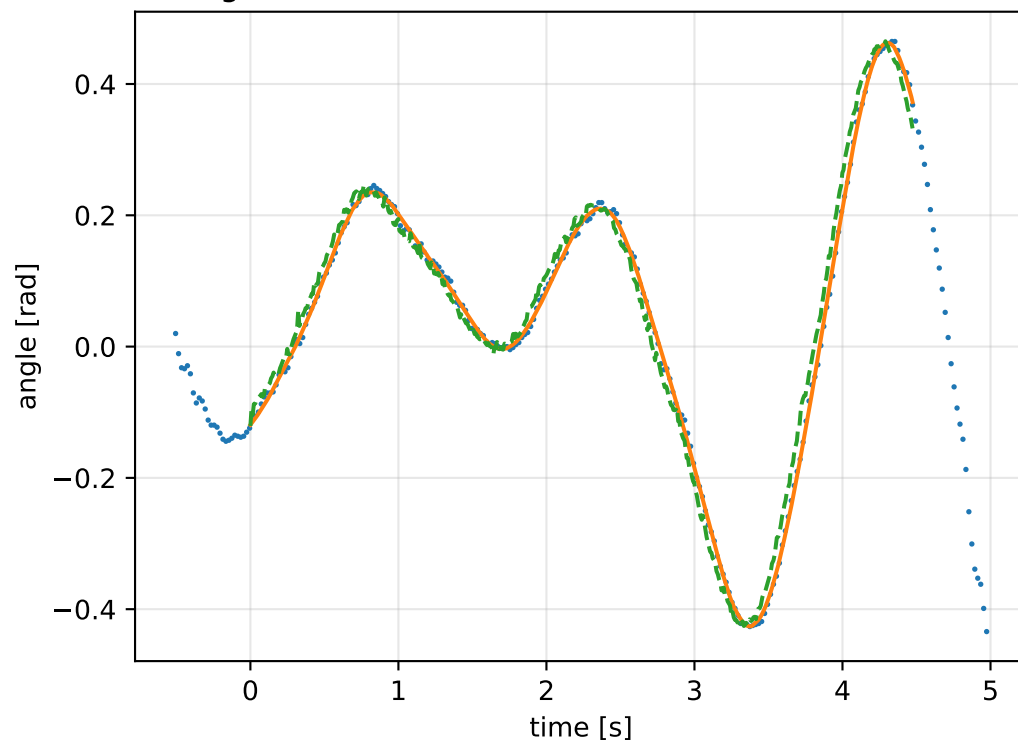
gimbal 2: RMSE=0.03107, max=0.08239 rad



gimbal 3: RMSE=0.03106, max=0.08682 rad

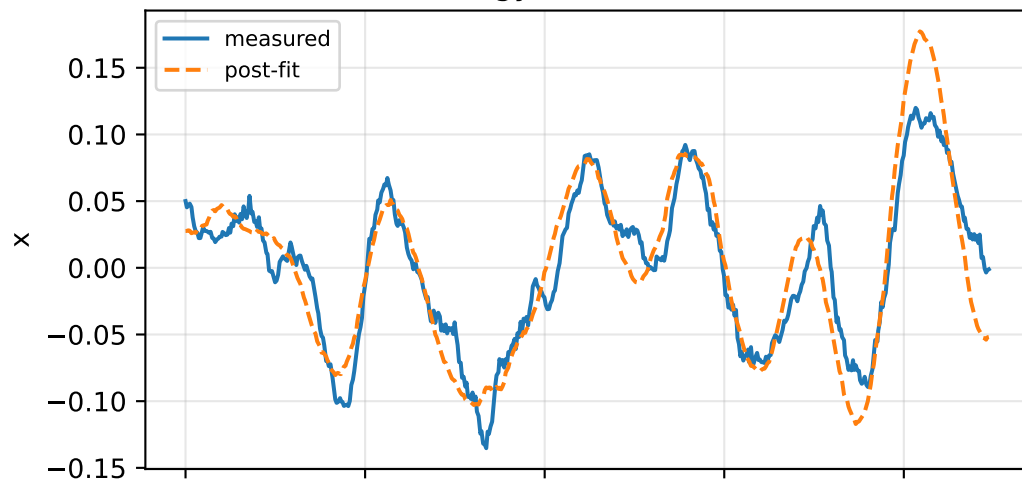


gimbal 4: RMSE=0.02598, max=0.07261 rad

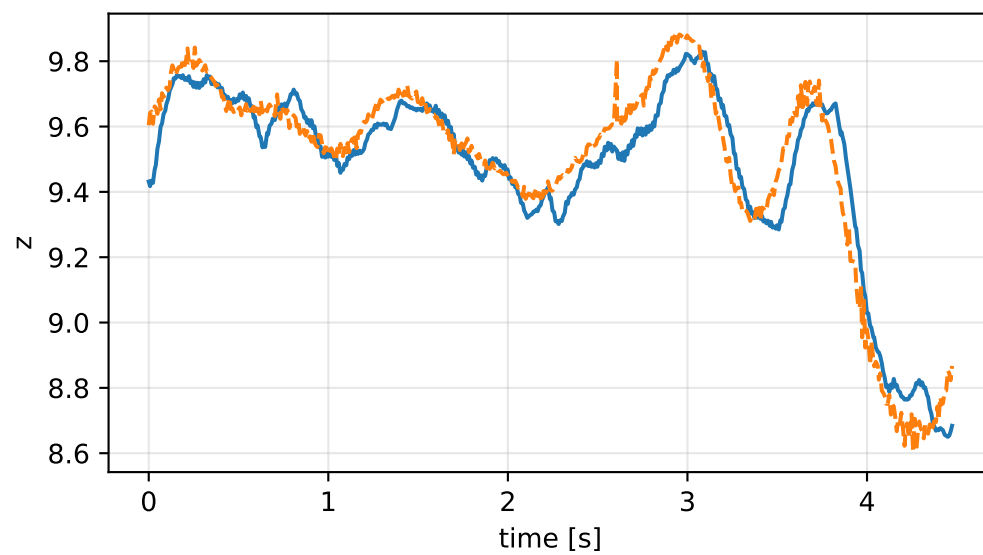
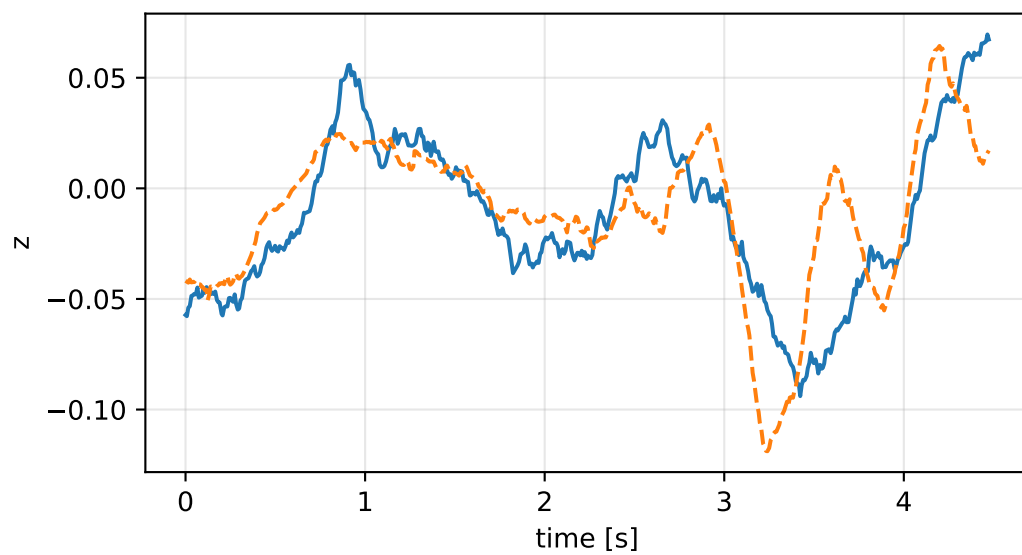
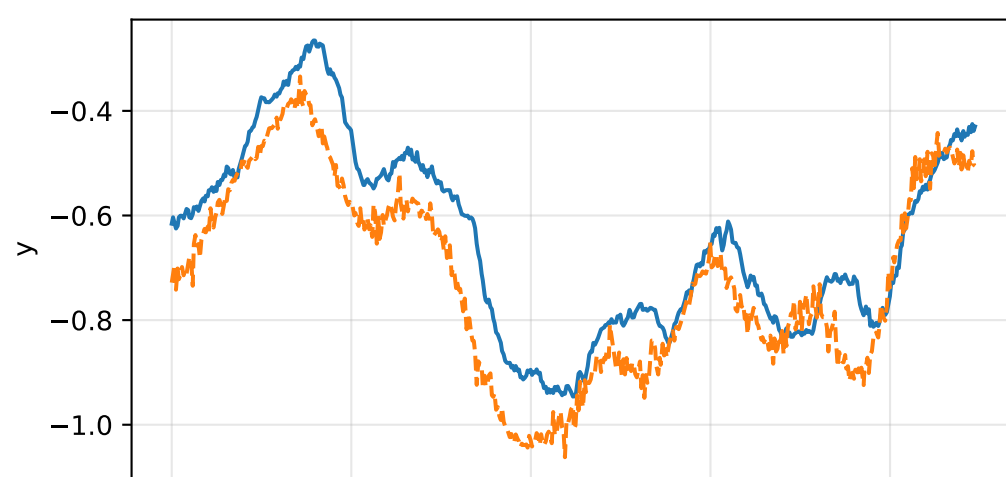
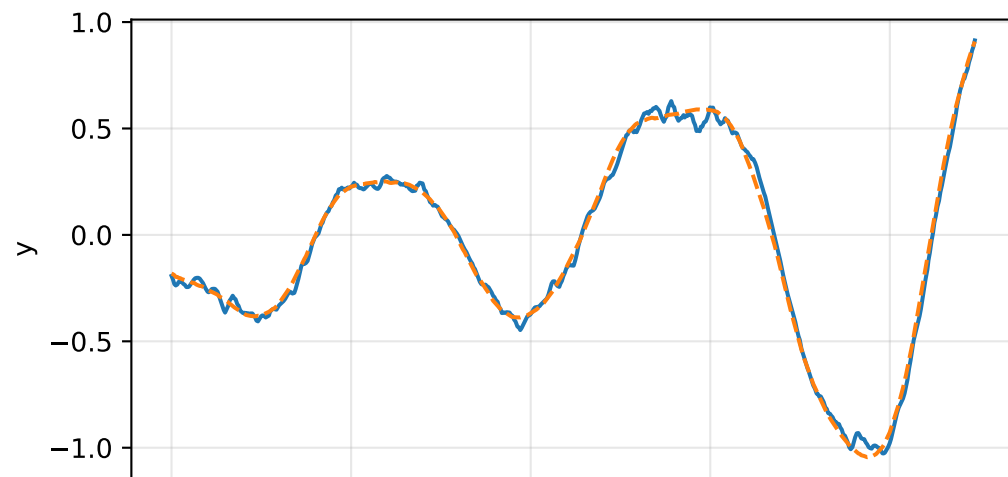
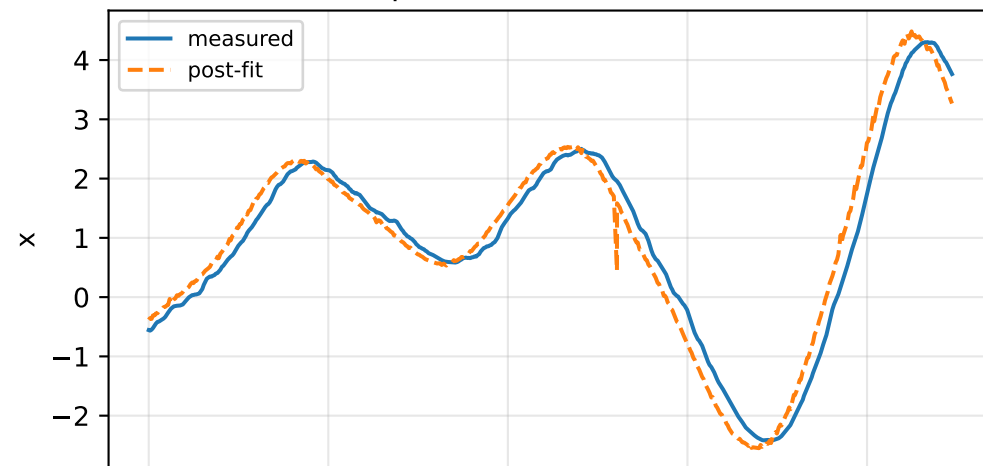


inertia_over_mass_yz_nominal: IMU comparison (diagnostic only)

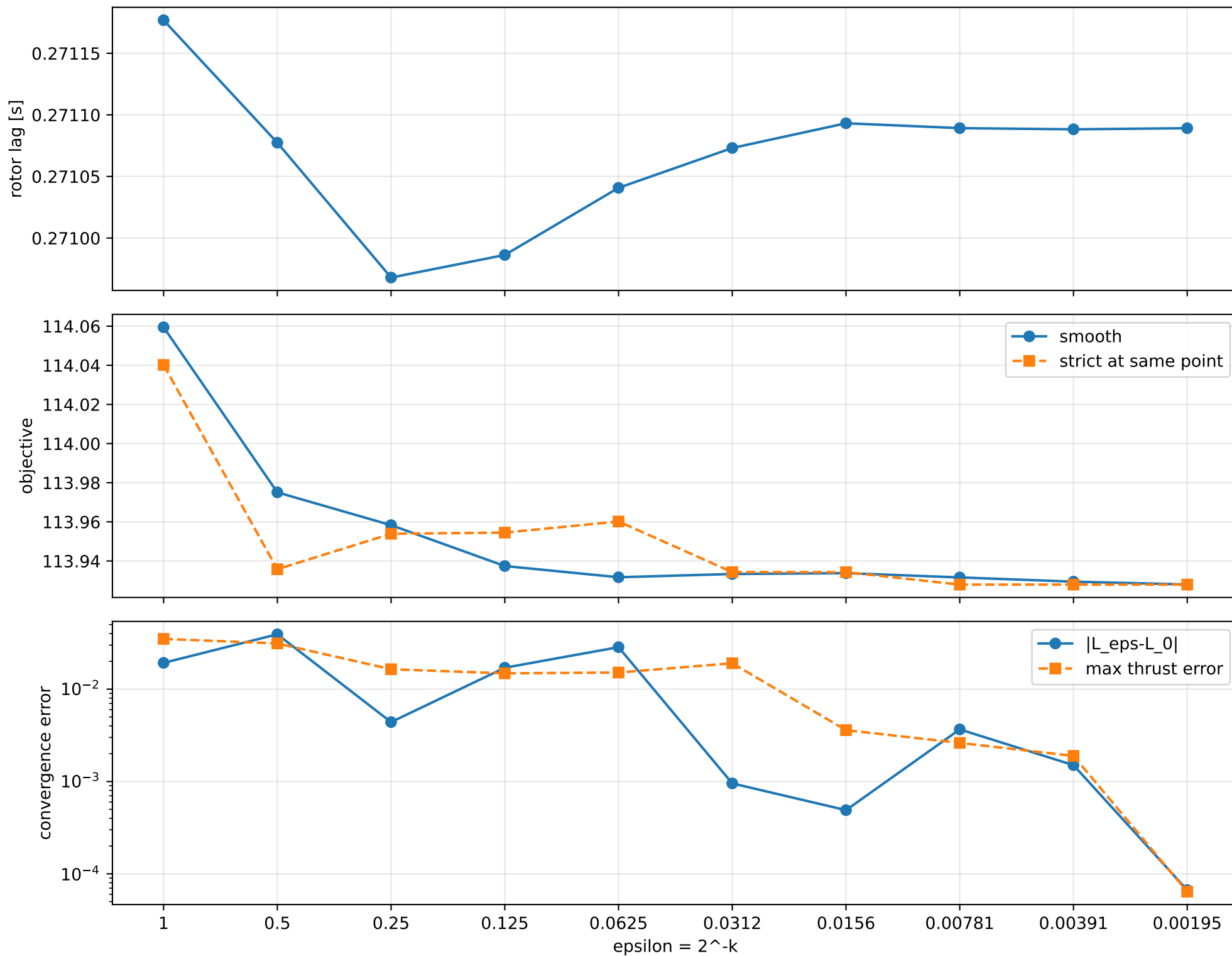
gyro [rad/s]



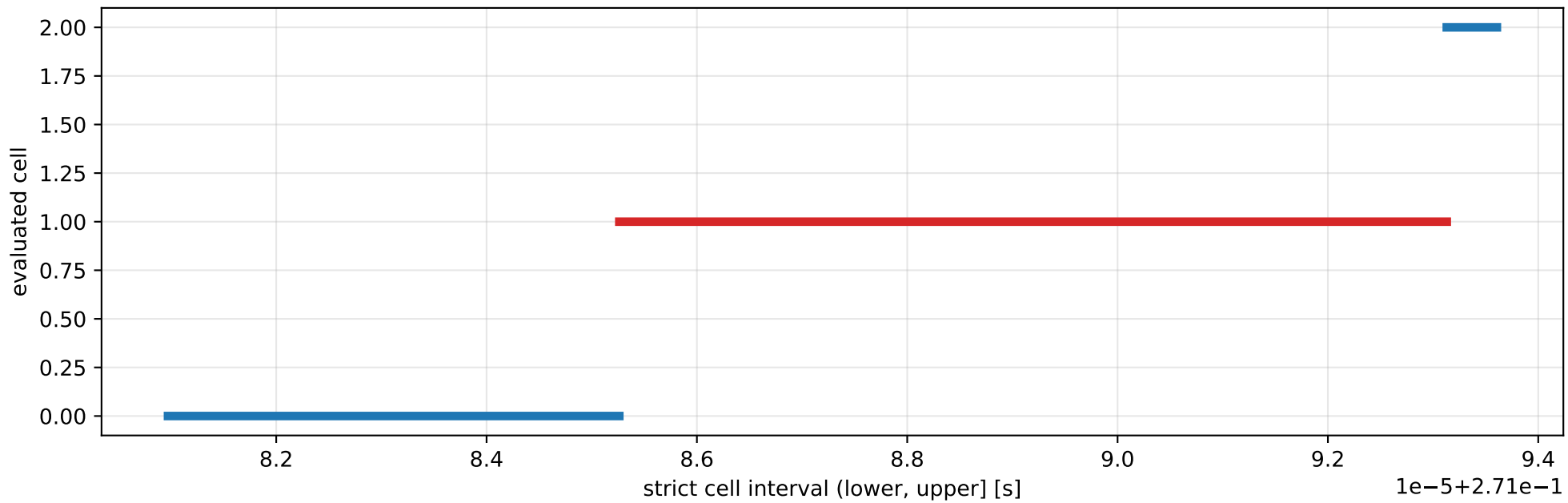
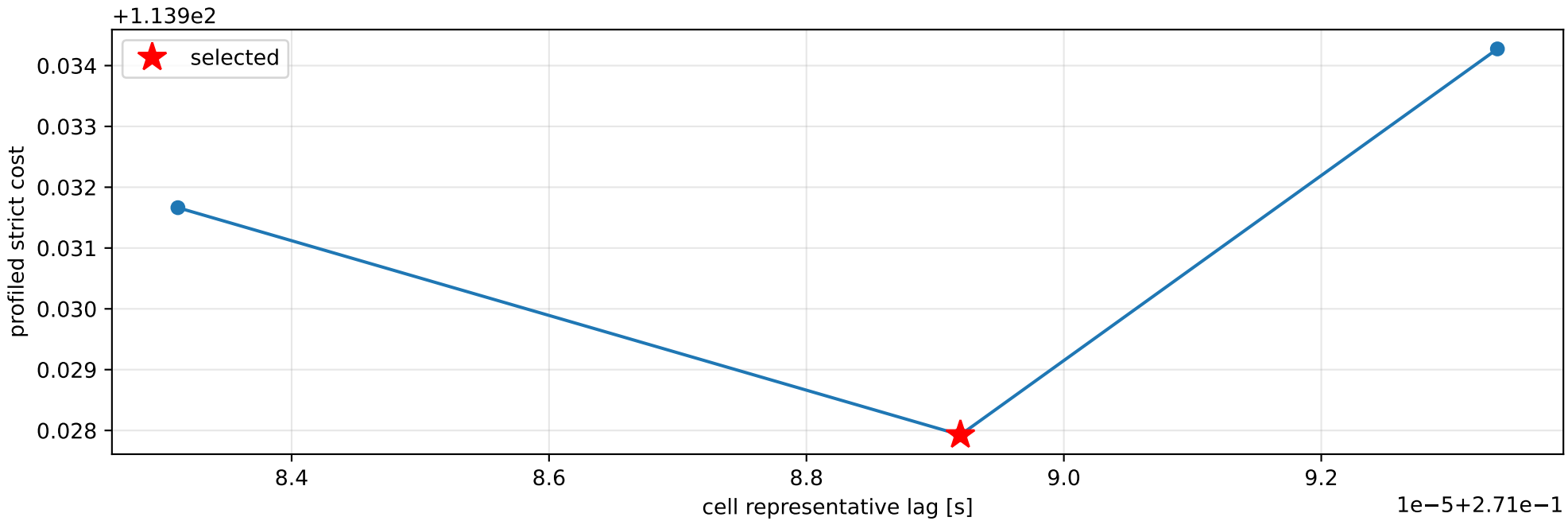
specific force [m/s^2]



inertia_over_mass_yz_nominal: smooth-to-strict continuation



inertia_over_mass_yz_nominal: exact strict-ZOH lag cells



selected (0.271085262, 0.27109313] s; final smooth 0.271089172 s; data boundary=False

inertia_over_mass_yz_nominal: parameter estimate

Case: inertia_over_mass_yz_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 4.75012514

inertia [kg m²]:

[[3.84345132e-01 1.21634239e-02 -2.98459549e-03]

[1.21634239e-02 1.58743784e-01 1.18545697e-07]

[-2.98459549e-03 1.18545697e-07 5.42972562e-01]]

CoG body [m]: [-0.02025038 0.00010397 0.0106116]

force effectiveness: [1.83728425 1.61709332 1.56761936 1.83167932]

scale-free J/m [m²]:

[[8.09126330e-02 2.56065336e-03 -6.28319340e-04]

[2.56065336e-03 3.34188635e-02 2.49563313e-08]

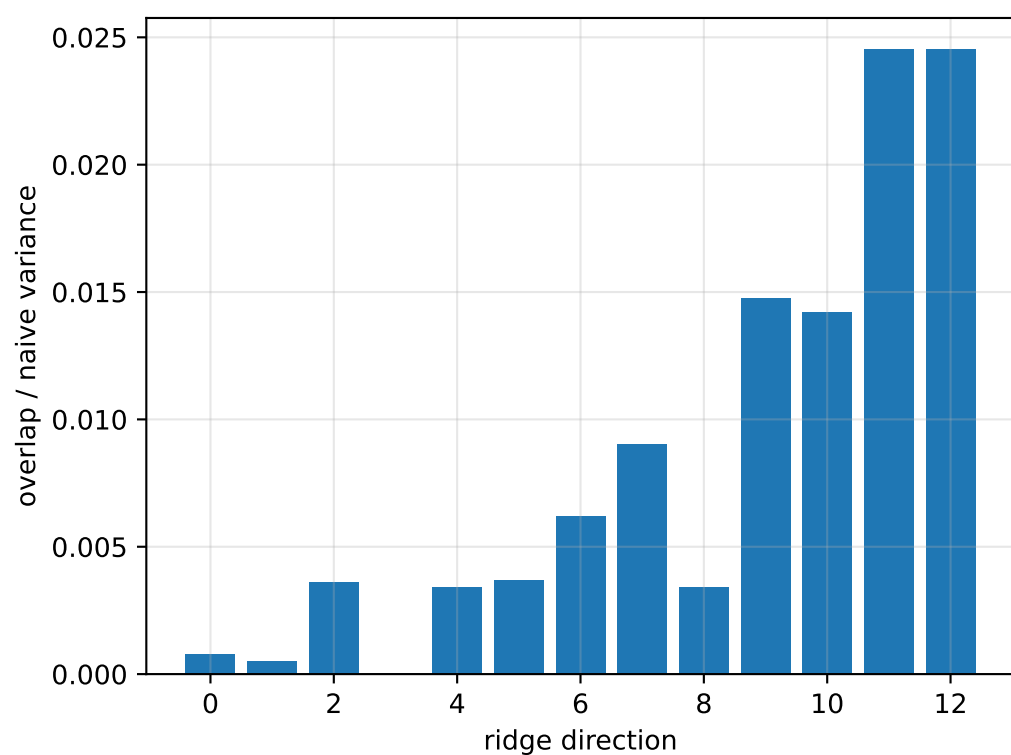
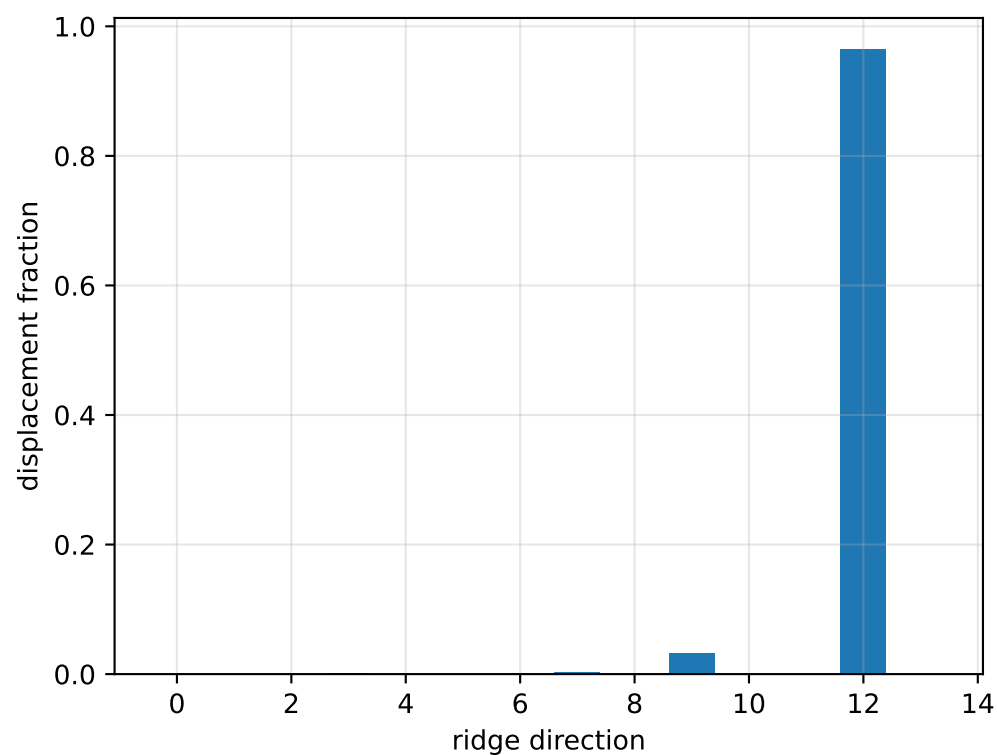
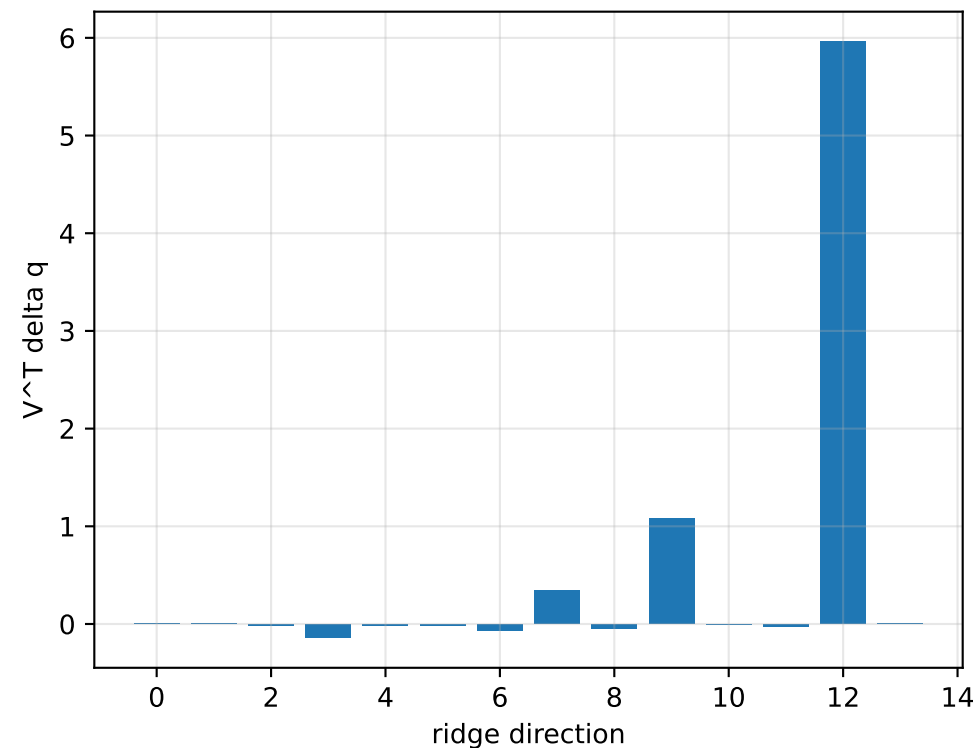
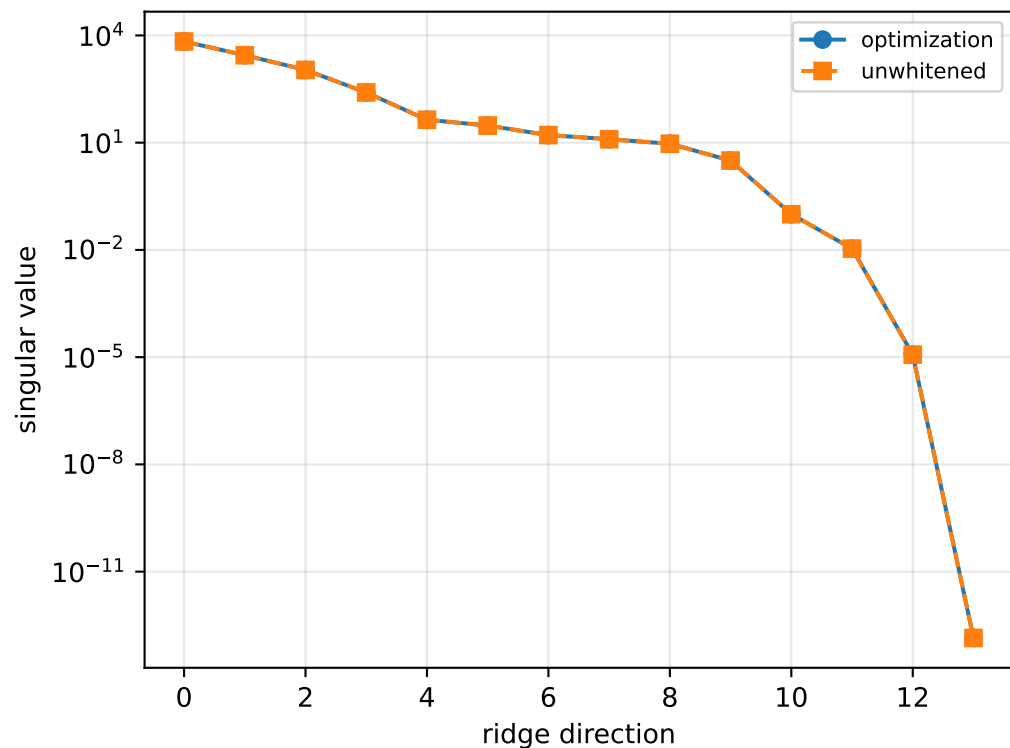
[-6.28319340e-04 2.49563313e-08 1.14307001e-01]]

scale-free f/m: [0.38678649 0.34043173 0.33001643 0.38560654]

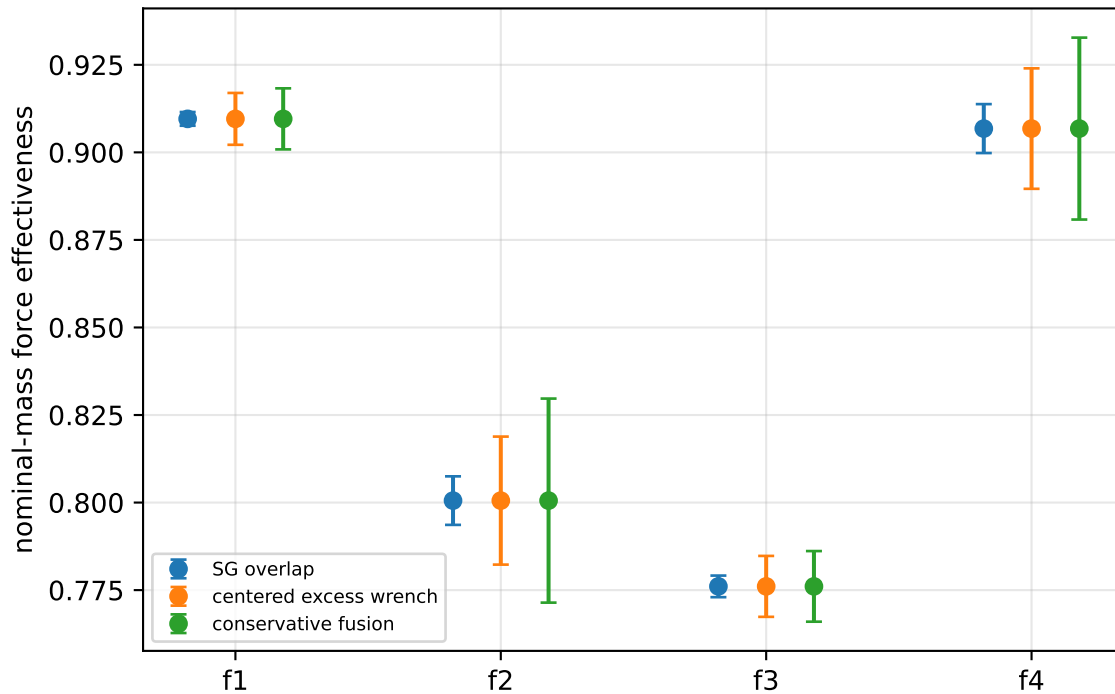
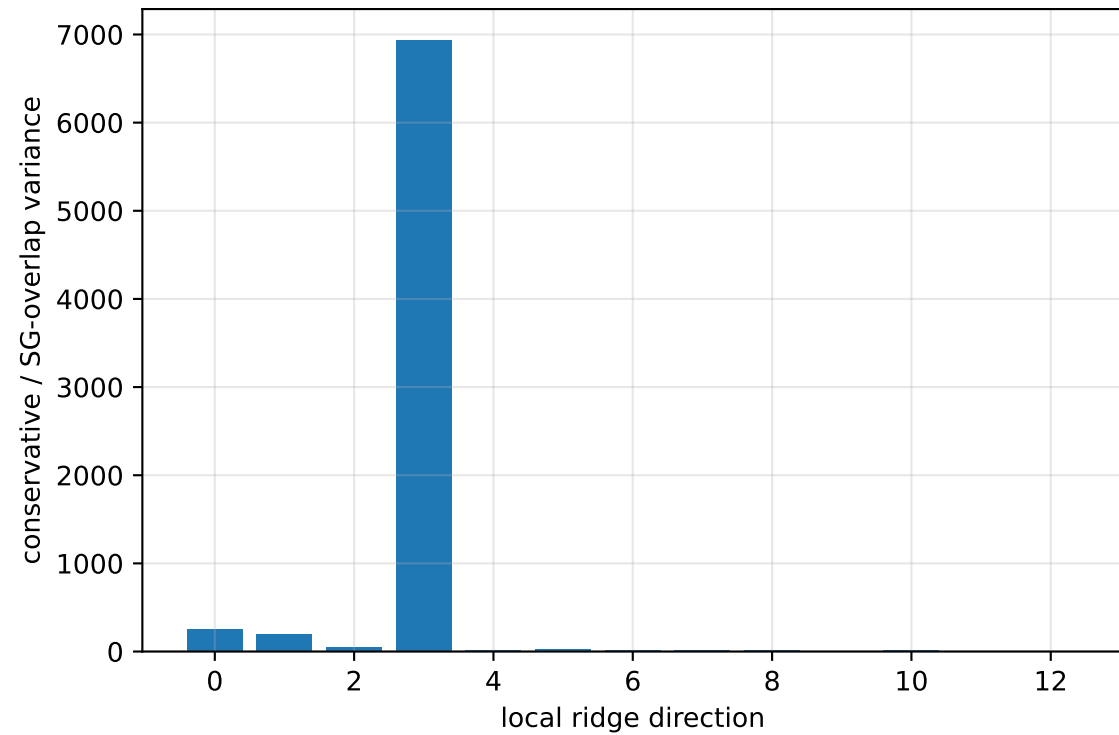
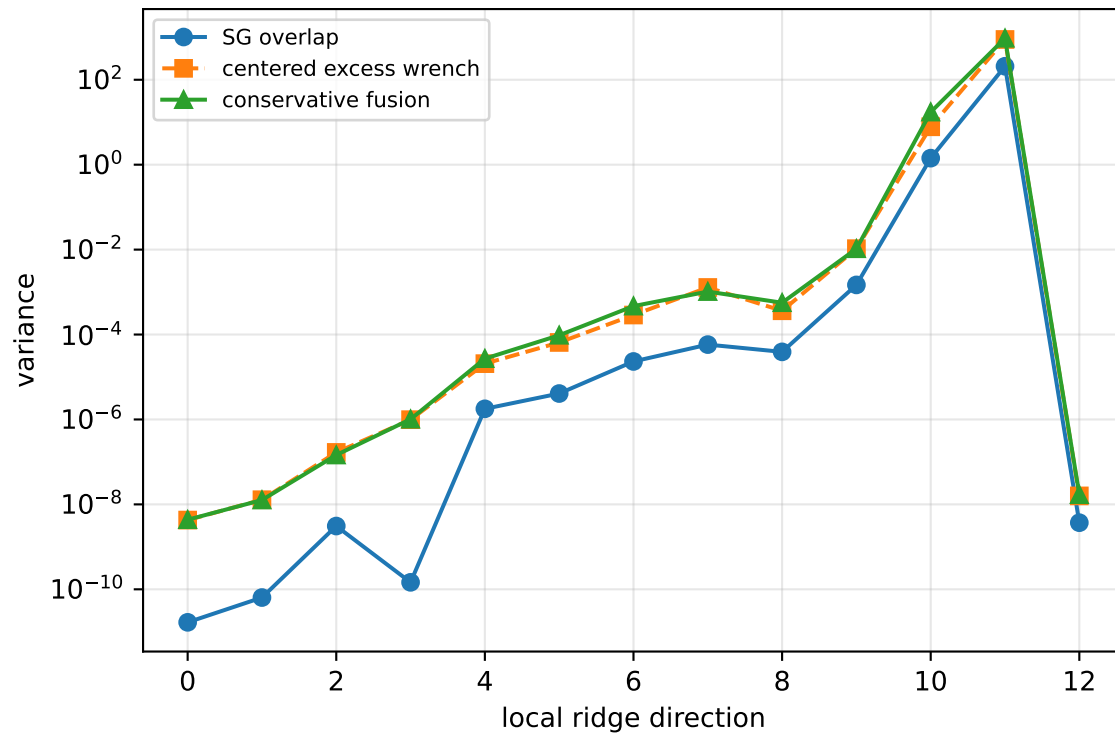
rotor lag cell: (0.271085262, 0.27109313] s

representative: 0.271089196 s

inertia_over_mass_yz_nominal: ridge spectrum (rank 13, nullity 1, $||\mathbf{v}||=1.5\text{e-}13$)



inertia_over_mass_yz_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 233.2$
wrench-to-acceleration recovery max error = $2.44\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

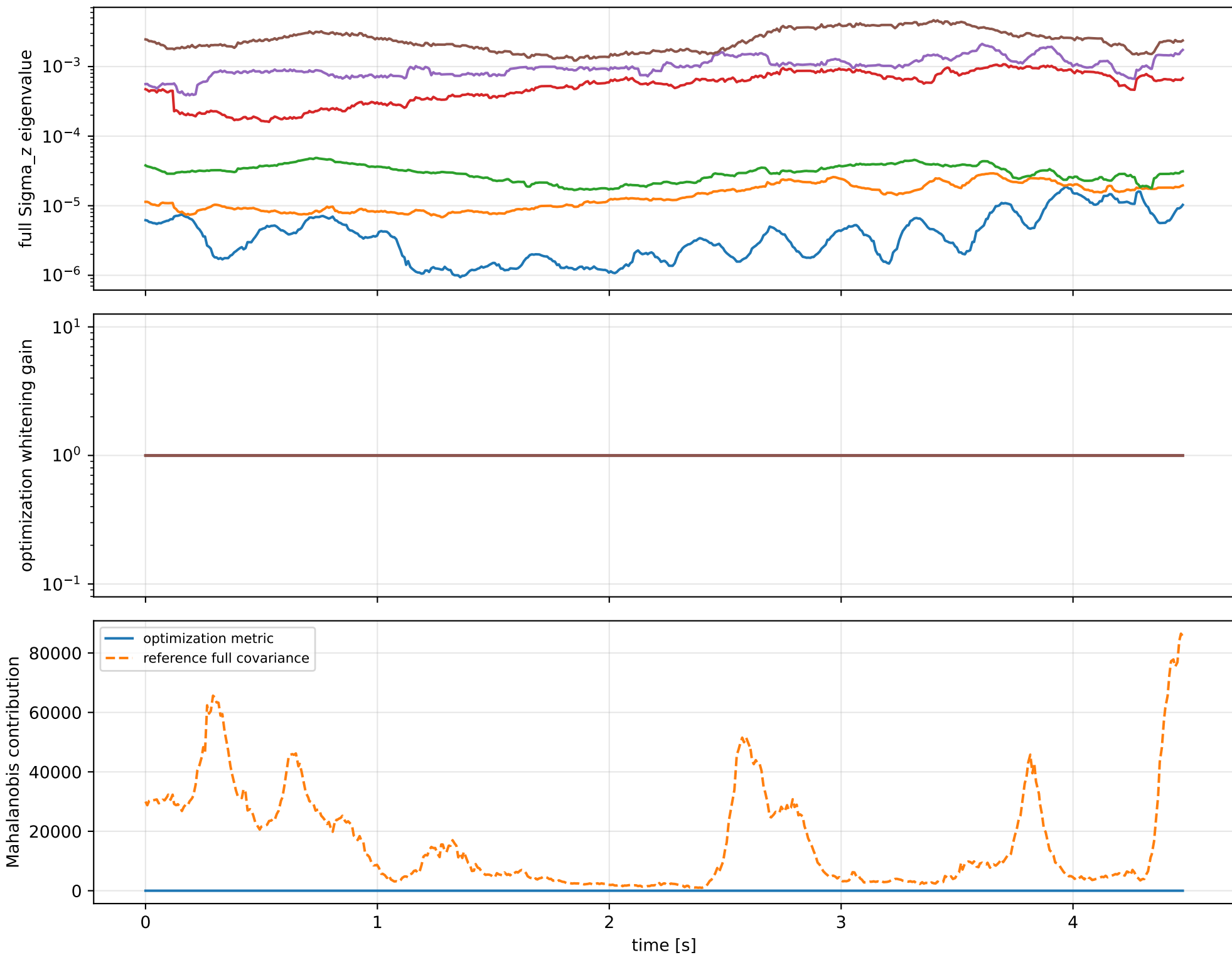
index 11: variance=948.7, ratio=4.561
 $\log_{\text{mass}}=+0.00621$; second_moment_diag_1=+0.259
second_moment_diag_2=+0.00761; second_moment_diag_3=-0.298
second_moment_offdiag_12=+0.0119; second_moment_offdiag_13=-0.918
second_moment_offdiag_23=-0.0326; cog_x=-1.29e-05
cog_y=-2.32e-06; cog_z=-1.09e-06
log_force_eff_1=+0.00626; log_force_eff_2=+0.00615
log_force_eff_3=+0.00614; log_force_eff_4=+0.00626

index 10: variance=17.31, ratio=12.14
 $\log_{\text{mass}}=+0.000643$; second_moment_diag_1=-0.00745
second_moment_diag_2=+0.00195; second_moment_diag_3=+0.00128
second_moment_offdiag_12=+0.172; second_moment_offdiag_13=+0.0347
second_moment_offdiag_23=-0.984; cog_x=+0.000575
cog_y=-0.000676; cog_z=-1.29e-05
log_force_eff_1=+0.000726; log_force_eff_2=+0.00707
log_force_eff_3=-0.000148; log_force_eff_4=-0.00408

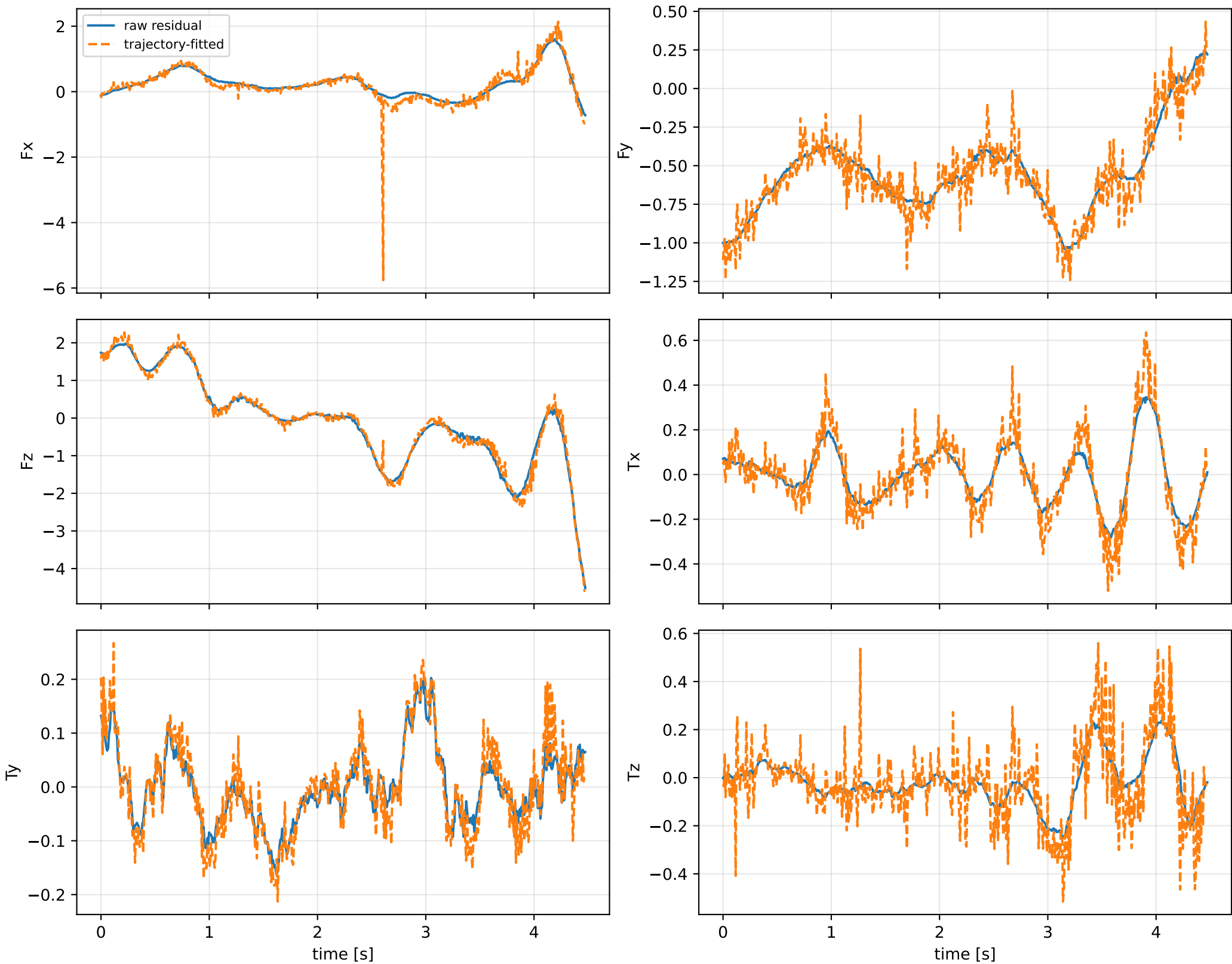
index 9: variance=0.01046, ratio=7.097
 $\log_{\text{mass}}=-0.151$; second_moment_diag_1=-0.145
second_moment_diag_2=+0.922; second_moment_diag_3=-0.00456
second_moment_offdiag_12=-0.00969; second_moment_offdiag_13=-0.0371
second_moment_offdiag_23=-0.00113; cog_x=-0.00976
cog_y=-0.00123; cog_z=-0.000668
log_force_eff_1=-0.108; log_force_eff_2=-0.19
log_force_eff_3=-0.209; log_force_eff_4=-0.114

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

inertia_over_mass_yz_nominal: optimization vs reference covariance



inertia_over_mass_yz_nominal: wrench / closure (force max $1.16\text{e-}14$, torque max $2.22\text{e-}16$)



inertia_over_mass_yz_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_inertia_over_mass_yz_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/inertia_over_ma

SHA256: cb359687f62672f99943f224e046f1d453f420c78c59be95963f7c1e6ba83846

data objective: 113.927927012

prior objective: 2.08366848561e-08

total objective: 113.927927033

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: inertia_over_mass_yz_nominal (inertia_over_mass_m2)

components: yz

target: [2.51604719e-08]

std: [1.e-06]

final: [2.49563313e-08]

error: [-2.04140564e-10]

standardized residual: [-0.00020414]

factor objective: 2.08366848561e-08